

01. What is Database

What is a Database?

- A **database** is an organised collection of data.
 - Stored electronically
 - Can be easily accessed, managed, and updated.
 - Databases are used to store information that businesses need to function efficiently.
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Key Concepts:

- **Data**: Information stored in the database (e.g., names, addresses, orders).
 - **Database Management System (DBMS)**: Software that allows for the creation, querying, updating, and management of databases (e.g., MySQL, PostgreSQL, MongoDB).
 - **Tables**: Structure within a relational database where data is organized in rows and columns (like a spreadsheet).
 - **Queries**: Requests made to the database to retrieve or manipulate data (e.g., SQL queries).
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Types of Databases:

- **Relational Databases**: Store data in structured tables with predefined relationships. Examples: MySQL, PostgreSQL, Oracle.
 - **NoSQL Databases**: Store data in non-tabular formats (key-value pairs, documents, etc.). Examples: MongoDB, Redis.
 - **Cloud Databases**: Hosted on cloud platforms, allowing for remote access and scalability. Examples: AWS RDS, Google Cloud SQL.
 - **Distributed Databases**: Spread over multiple locations, improving accessibility and redundancy. Examples: Apache Cassandra, Google Bigtable.
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How Databases Work

- **Data Insertion**: Data is added to tables using queries.
- **Data Retrieval**: Users or applications can retrieve specific data by running queries.
- **Data Updates**: Existing records can be modified through update commands.
- **Data Deletion**: Unwanted data can be removed by executing delete commands.

```
INSERT INTO contacts (name, email, phone, subject, msg, status)
VALUES ('Rahim Uddin', 'rahim@example.com', '01700000000', 'Inquiry', 'I want more information', 1);

SELECT * FROM contacts;

UPDATE contacts
SET subject = 'Updated Subject',
    msg = 'Updated message'
WHERE status = 1;

DELETE FROM contacts
WHERE status = 1;
```

Why are Databases Important

- **Efficiency:** Databases allow businesses to handle large volumes of data efficiently.
- **Accuracy:** Well-designed databases ensure accurate and consistent data management.
- **Security:** Modern databases provide access controls to secure sensitive data.
- **Scalability:** Databases can grow as the business grows, handling more users and data over time.